
FINITE PRESENTATIONS OF METABELIAN GROUPS: EFFECTIVE ENUMERATION VIA LAURENT RELATIONS

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ABSTRACT

For an ordinary finite presentation $P = \langle x_1, \dots, x_n \mid R \rangle$, put $G(P) = F_n / \langle\langle R \rangle\rangle$. We construct a primitive-recursive predicate V with

$$G(P)'' = 1 \iff \exists c \in \mathbb{N} : V(P, c) = 1.$$

Thus finite presentations of metabelian groups are recursively enumerable, answering Kourovka Problem 17.124. An effective form of the Bieri–Strebel covering construction, using signed Laurent relations and rational separation, gives a family of finitely presented metabelian groups cofinal under epimorphisms. Products of conjugates of defining relators witness these epimorphisms. The construction and enumeration theorem are formalized in Lean 4.

1 The enumeration theorem

Let $F_n = F(x_1, \dots, x_n)$ and let R be a finite list of words in F_n . We consider ordinary presentations

$$P = \langle x_1, \dots, x_n \mid R \rangle, \quad G(P) = F_n / \langle\langle R \rangle\rangle, \quad \mathcal{M} = \{P : G(P)'' = 1\}.$$

The empty generating set is allowed. Encodings with a letter outside $\{x_1, \dots, x_n\}$ are rejected.

Theorem 1.1. *There is a primitive-recursive predicate V on natural-number encodings of presentations and finite witnesses such that*

$$P \in \mathcal{M} \iff \exists c \in \mathbb{N} : V(P, c) = 1. \tag{1.1}$$

In particular, $\mathcal{M}$ is recursively enumerable.

This answers the enumeration question in Kourovka Problem 17.124 [5] and the enumeration and witness questions of Shpilrain [7, Problems 4 and 5]. The ordinary-presentation recognition question also occurs in [2, Problem 42, arXiv version], with attribution to Miller [6].

The covering presentations come from Bieri and Strebel [3, §3.5, equations (3.1)–(3.4), Lemma 3.5 and §3.6]. We make their finite data effectively recognizable and obtain ordinary finite presentations P_R satisfying

$$P_R'' = 1, \quad G'' = 1, \quad G \text{ finitely presented} \implies \exists P_R \twoheadrightarrow G. \tag{1.2}$$

A finite witness for the epimorphism consists of identities in a free group, each a product of conjugates of defining relators. These two ingredients give (1.1).

The algorithms of Baumslag, Cannonito and Robinson [1, p. 630 and Theorems 2.1, 3.1] take finite presentations in the metabelian variety. Here the input is $F_n/\langle\langle R \rangle\rangle$ in the variety of all groups. The geometric and module-theoretic ingredients remain classical; the construction makes their finite data sufficient for ordinary-presentation enumeration.

For abelian-by-cyclic groups, recursive enumerability of ordinary finite presentations is already recorded in the proof of [4, Lemma 11.4]. Theorem 1.1 treats arbitrary metabelian groups. Its certificate gives explicit finite Laurent data and normal-closure derivations witnessing metabelianity.

2 Conventions and classical inputs

We use right conjugation and right commutators:

$$a^w = w^{-1}aw, \quad [a, b] = a^{-1}b^{-1}ab.$$

A group G is metabelian when its derived subgroup G' is abelian, equivalently when $[[a, b], [c, d]] = 1$ for all $a, b, c, d \in G$. Given letters $t_1, \dots, t_k$, put

$$q(v) = t_1^{v_1} \cdots t_k^{v_k} \quad (v \in \mathbb{Z}^k).$$

We retain the literal inverse

$$q(v)^{-1} = t_k^{-v_k} \cdots t_1^{-v_1}.$$

Unsubscripted norms are Euclidean.

Lemma 2.1 (Bieri–Strebel collection). *Let $k \geq 1$ and let $\mathcal{A}$ be a finite alphabet, with a designated letter a_{ij} for each $1 \leq i < j \leq k$. For $\rho > 0$, let H_ρ have generators $t_1, \dots, t_k$ and $\mathcal{A}$, and relators*

$$[t_i, t_j] = a_{ij}, \quad [a, b^{q(v)}] = 1 \quad (a, b \in \mathcal{A}, v \in \mathbb{Z}^k, \|v\| < \rho).$$

Suppose $\rho > 2k$ and $u, v \in \mathbb{Z}^k$ satisfy

$$\|u\| \leq \frac{\rho}{2k}, \quad \|v\| < \rho + \frac{1}{2k}, \quad \|u + v\| < \rho. \quad (2.1)$$

Then, for all $a, b \in \mathcal{A}$, the following hold in H_ρ :

$$[a, b^{q(u)q(v)}] = 1, \quad [a, b^{q(u)^{-1}q(v)^{-1}}] = 1.$$

If all ordered commutation relators are imposed, with no radius restriction, the normal closure of $\mathcal{A}$ is abelian.

These are the ordered and inverse-ordered cases of [3, Lemma 3.4b]; the normal-closure assertion is [3, Corollary 3.3]. The collection argument [3, Lemma 3.2] is valid in every quotient of H_ρ .

For the second input, let A be an abelian normal subgroup of a group E with $E/A \cong \mathbb{Z}^k$. Conjugation gives A a right module structure over $\mathbb{Z}[t_1^{\pm 1}, \dots, t_k^{\pm 1}]$, independent of the chosen lifts of the quotient generators. Let A^* denote the inverse-action module, in which t_i acts as t_i^{-1} acts on A .

Lemma 2.2 (Finite signed tameness data). *Let $k \geq 1$. Suppose E is finitely presented and metabelian, $E/A \cong \mathbb{Z}^k$, and A is finitely generated as a $\mathbb{Z}[t_1^{\pm 1}, \dots, t_k^{\pm 1}]$ -module. There is a finite family of nonzero Laurent polynomials*

$$\lambda = \sum_{u \in S_\lambda} c_{\lambda,u} t^u \quad (c_{\lambda,u} \in \mathbb{Z} \setminus \{0\}),$$

indexed by a signed family $\Lambda = \Lambda_+ \sqcup \Lambda_-$, such that

$$A(\lambda - 1) = 0 \quad (\lambda \in \Lambda_+), \quad A^*(\lambda - 1) = 0 \quad (\lambda \in \Lambda_-),$$

$$\forall v \in \mathbb{R}^k \setminus \{0\} \exists \lambda \in \Lambda : \min_{u \in S_\lambda} v \cdot u > 0.$$

Apply the necessity direction of [3, Theorem A] and the centralizer criterion [3, Lemma 2.6] to A and A^* . For $A = 0$, use the monomials $t_i^{\pm 1}$; for $A \neq 0$, an identity-acting polynomial is nonzero. The minus sign changes the module action; the displayed support remains S_λ .

3 Rational separation and the commutation radius

Fix $k \geq 1$ and a finite family $(S_\lambda)_{\lambda \in \Lambda}$ of nonempty subsets of $\mathbb{Z}^k$. Consider

$$\forall v \neq 0 \exists \lambda \forall u \in S_\lambda : v \cdot u > 0. \quad (3.1)$$

Only rational arithmetic is needed to test this condition.

Lemma 3.1 (Rational separation). *Condition (3.1) is decidable. If it holds, there is an integer $r \geq 0$ such that, for $\varepsilon = 2^{-r}$,*

$$\forall v \in \mathbb{R}^k, \|v\|_\infty = 1 \implies \exists \lambda \forall u \in S_\lambda : v \cdot u > \varepsilon. \quad (3.2)$$

Whether a proposed r satisfies (3.2) is decidable by a primitive-recursive computation on the finite integer data.

Proof. Put $\Sigma = \{v \in \mathbb{R}^k : \|v\|_\infty = 1\}$. Its faces are

$$F_{j,s} = \{v \in [-1, 1]^k : v_j = s\}, \quad \Sigma = \bigcup_{j=1}^k \bigcup_{s \in \{-1, 1\}} F_{j,s}.$$

Failure of (3.2) on $F_{j,s}$ is equivalent to the existence of $(u_\lambda)_\lambda \in \prod_\lambda S_\lambda$ and v satisfying

$$v \cdot u_\lambda \leq \varepsilon \quad (\lambda \in \Lambda), \quad -1 \leq v_i \leq 1, \quad v_j = s. \quad (3.3)$$

Fourier–Motzkin elimination is the following rational equivalence. For $\alpha_i x + \beta_i \cdot y \leq \gamma_i$, put $I_\pm = \{i : \pm \alpha_i > 0\}$ and $I_0 = \{i : \alpha_i = 0\}$. Then

$$\exists x \forall i : \alpha_i x + \beta_i \cdot y \leq \gamma_i \iff \begin{cases} \beta_i \cdot y \leq \gamma_i & (i \in I_0), \\ \frac{\gamma_n - \beta_n \cdot y}{\alpha_n} \leq \frac{\gamma_p - \beta_p \cdot y}{\alpha_p} & (n \in I_-, p \in I_+). \end{cases}$$

After k eliminations, only rational inequalities remain. At each stage, rational values of the remaining coordinates admit a rational choice of the eliminated coordinate. Thus rational and real feasibility agree. Applying the same test with $\varepsilon = 0$ decides (3.1).

Under (3.1), $\Lambda \neq \emptyset$ and compactness gives

$$0 < \mu := \min_{v \in \Sigma} \max_{\lambda} \min_{u \in S_{\lambda}} v \cdot u, \quad 2^{-r} < \mu \implies (3.2).$$

Hence some $r \in \mathbb{N}$ succeeds. For a supplied r , the k elimination steps, finite choices of supports and exact rational comparisons are primitive recursive. $\square$

For data passing this margin test put

$$C = \frac{2^{-r}}{k}, \quad D = 1 + \sum_{\lambda} \sum_{u \in S_{\lambda}} \|u\|_1, \quad R = 1 + 2k(1 + D + D^2 k 2^r). \quad (3.4)$$

Thus D is an integer, $0 < C \leq 1$, and $R = 1 + 2k(1 + D + D^2/C)$. In particular,

$$R > 2k, \quad D < \frac{R}{2k}, \quad R > \max\{C, D^2/C\}. \quad (3.5)$$

By homogeneity and $\|v\|_{\infty} \geq 1/k$ for $\|v\| = 1$, (3.2) implies that for each Euclidean unit vector v some support satisfies $v \cdot u > C$ throughout.

Lemma 3.2 (Radial propagation). *Put $\delta = \min\{C/4, 1/(4k)\}$. If $\rho \geq R$ and $\rho \leq \|v\| < \rho + \delta$, there is a support S_{λ} such that*

$$\|v + u\| < \rho \quad (u \in S_{\lambda}).$$

Proof. Put $s = \|v\|$. The margin in direction $-v/\|v\|$ gives λ with $v \cdot u \leq -Cs$ for all $u \in S_{\lambda}$. For each such u , $\|u\| < D$, and

$$\begin{aligned} \|v + u\|^2 &= s^2 + 2v \cdot u + \|u\|^2 \\ &< s^2 - 2Cs + D^2 \\ &< (\rho + \delta)^2 - 2C(\rho + \delta) + D^2 \\ &= \rho^2 - 2(C - \delta)\rho + \delta^2 - 2C\delta + D^2 \\ &\leq \rho^2 - \frac{3}{2}C\rho + D^2 < \rho^2. \end{aligned}$$

Indeed $s \geq \rho \geq R > C$, $\delta \leq C/4$ and $\rho > D^2/C$. The same λ works for every $u \in S_{\lambda}$. $\square$

4 Metabelian presentations from Laurent relations

A *cover datum* consists of $k \geq 1$, a finite nonempty alphabet $\mathcal{A}$ with designated letters a_{ij} , a finite family of signed nonzero Laurent polynomials, and an integer r passing the margin test. Write Λ_+ and Λ_- for the two signed families, and fix an order on every support. Using the radius (3.4), define

P_R on the alphabet $\{t_1, \dots, t_k\} \cup \mathcal{A}$ by

$$[t_i, t_j] = a_{ij} \quad (1 \leq i < j \leq k), \quad (4.1)$$

$$[a, b^{q(v)}] = 1 \quad (a, b \in \mathcal{A}, \|v\| < R), \quad (4.2)$$

$$a = \prod_{u \in S_\lambda} (a^{c_{\lambda,u}})^{q(u)} \quad (a \in \mathcal{A}, \lambda \in \Lambda_+), \quad (4.3)$$

$$a = \prod_{u \in S_\lambda} (a^{c_{\lambda,u}})^{q(u)^{-1}} \quad (a \in \mathcal{A}, \lambda \in \Lambda_-). \quad (4.4)$$

Products follow the chosen support order; the minus case uses the literal inverse $q(u)^{-1}$. The relator set is finite since

$$B_R = \{v \in \mathbb{Z}^k : \|v\| < R\} = \left\{ v \in [-R, R]^k \cap \mathbb{Z}^k : \sum_i v_i^2 < R^2 \right\}.$$

Proposition 4.1 (Metabelianity). *Every group P_R constructed from an accepted cover datum is metabelian.*

Proof. In a group satisfying (4.1), (4.3) and (4.4), write

$$C(\rho) : [a, b^{q(v)}] = 1 \quad (a, b \in \mathcal{A}, v \in \mathbb{Z}^k, \|v\| < \rho).$$

We prove $C(\rho) \Rightarrow C(\rho + \delta)$ for $\rho \geq R$.

Only vectors v with $\rho \leq \|v\| < \rho + \delta$ need consideration. Choose λ by Lemma 3.2. Then, for every $u \in S_\lambda$,

$$\|u\| < D < \rho/(2k), \quad \|v\| < \rho + 1/(2k), \quad \|u + v\| < \rho. \quad (4.5)$$

If λ has sign $+$, Lemma 2.1 and (4.5) give

$$[a, b^{q(u)q(v)}] = 1 \quad (u \in S_\lambda).$$

Writing $C_H(a)$ for the centralizer of a in the ambient group, relation (4.3) therefore yields

$$b^{q(v)} = \prod_{u \in S_\lambda} (b^{c_{\lambda,u}})^{q(u)q(v)} \in C_H(a).$$

If λ has sign $-$, the inverse-word assertion of Lemma 2.1, with the alphabet letters interchanged, gives

$$[b, a^{q(u)^{-1}q(v)^{-1}}] = 1 \quad (u \in S_\lambda).$$

Now (4.4) yields

$$a^{q(v)^{-1}} = \prod_{u \in S_\lambda} (a^{c_{\lambda,u}})^{q(u)^{-1}q(v)^{-1}} \in C_H(b).$$

Conjugating this commutation identity by $q(v)$ gives $[a, b^{q(v)}] = 1$. Thus both signs prove the propagation step.

In P_R , relation (4.2) supplies $C(R)$, so

$$C(R + n\delta) \quad (n \in \mathbb{N}), \quad \bigcup_{n \geq 0} \{v \in \mathbb{Z}^k : \|v\| < R + n\delta\} = \mathbb{Z}^k.$$

All ordered commutators vanish. Lemma 2.1 and (4.1) imply, for $A_R = \langle\langle \mathcal{A} \rangle\rangle$,

$$A'_R = 1, \quad (P_R/A_R)' = 1, \quad P'_R \leq A_R, \quad P''_R = 1. \quad \square$$

5 Cofinality of the covers

Lemma 5.1 (Central extensions). *If $1 \rightarrow K \rightarrow E \rightarrow G \rightarrow 1$ is a central extension, G is finitely presented, and K is free abelian of finite rank, then E is finitely presented.*

Proof. Let $G = \langle x_1, \dots, x_m \mid r_1, \dots, r_\ell \rangle$. Choose lifts $y_i \in E$ and a basis $z_1, \dots, z_s$ of K ; write

$$r_j(y) = \prod_{h=1}^s z_h^{b_{jh}}, \quad b_{jh} \in \mathbb{Z}.$$

For alphabets $Y = \{y_1, \dots, y_m\}$ and $Z = \{z_1, \dots, z_s\}$, set

$$\begin{aligned} \mathcal{R}_0 &= \{[z_h, z_{h'}] : 1 \leq h < h' \leq s\} \cup \{[z_h, y_i] : 1 \leq h \leq s, 1 \leq i \leq m\}, \\ \mathcal{R}_1 &= \left\{ r_j(y) \left(\prod_{h=1}^s z_h^{b_{jh}} \right)^{-1} : 1 \leq j \leq \ell \right\}. \end{aligned}$$

The lifts induce an epimorphism

$$\varphi : \tilde{E} := \langle Y \sqcup Z \mid \mathcal{R}_0 \cup \mathcal{R}_1 \rangle \twoheadrightarrow E.$$

If $w \in \ker \varphi$, its image in G is trivial. Express its Y -word as a product of conjugates of $r_j^{\pm 1}$. The relations $\mathcal{R}_1$ and centrality relations $\mathcal{R}_0$ give

$$w = \prod_{h=1}^s z_h^{n_h} \quad \text{in } \tilde{E}, \quad 1 = \varphi(w) = \prod_{h=1}^s z_h^{n_h} \quad \text{in } K.$$

Independence of the basis implies $n_1 = \dots = n_s = 0$. Hence $\ker \varphi = 1$ and $\tilde{E} \cong E$. $\square$

Proposition 5.2 (Quotient cofinality). *Every finitely presented metabelian group G is an epimorphic image of some certified cover P_R .*

Proof. Put $A = G'$ and $Q = G/A$. Choose a surjection $\pi : \mathbb{Z}^k \twoheadrightarrow Q$ with $k \geq 1$ and form the pullback

$$E = \{(g, z) \in G \times \mathbb{Z}^k : gA = \pi(z)\}.$$

Both projections are surjective. Their kernels give exact sequences

$$1 \longrightarrow A \longrightarrow E \longrightarrow \mathbb{Z}^k \longrightarrow 1, \quad 1 \longrightarrow \ker \pi \longrightarrow E \longrightarrow G \longrightarrow 1. \quad (5.1)$$

$$\begin{array}{ccccc}
 P_R & \longrightarrow & E & \xrightarrow{\text{pr}_1} & G \\
 & & \downarrow \text{pr}_2 & & \downarrow \\
 & & \mathbb{Z}^k & \xrightarrow{\pi} & Q = G/G'
 \end{array}$$

Figure 1: The pullback $E = G \times_Q \mathbb{Z}^k$ and the epimorphism from P_R . All arrows are surjective.

The second kernel consists of $(1, z)$ with $z \in \ker \pi$, and is central and free abelian of finite rank. Lemma 5.1 makes E finitely presented. As a subgroup of the metabelian group $G \times \mathbb{Z}^k$, it is metabelian.

For a finite generating set $g_1, \dots, g_m$ of G ,

$$G' = \langle\langle [g_i, g_j] : i < j \rangle\rangle, \quad A = \sum_{i < j} [g_i, g_j] \mathbb{Z}Q.$$

The second equality uses additive module notation for the abelian group A . Pullback along π preserves these module generators.

Choose lifts $t_i \in E$ of the standard basis of $\mathbb{Z}^k$. Take a finite module-generating alphabet $\mathcal{A}$ for A , including letters representing every $[t_i, t_j]$. A letter representing the identity may be added to ensure that the alphabet is nonempty. Lemma 2.2 supplies signed Laurent identities on this module. Lemma 3.1 supplies a finite margin certificate.

The selected elements satisfy (4.1) by construction, (4.2) because $A' = 1$, and (4.3)–(4.4) by the module identities. They induce $\psi : P_R \rightarrow E$. If B is the image of $\mathcal{A}$, then

$$A = \langle b^{q(v)} : b \in B, v \in \mathbb{Z}^k \rangle \leq \text{im } \psi, \quad E = A \langle t_1, \dots, t_k \rangle = \text{im } \psi.$$

Thus ψ is surjective, and $P_R \twoheadrightarrow E \twoheadrightarrow G$. □

6 Epimorphisms and normal closures

For $P = \langle X \mid R \rangle$ and $w \in F(X)$,

$$w \in \langle\langle R \rangle\rangle \iff w = \prod_{v=1}^m h_v^{-1} r_{j_v}^{\epsilon_v} h_v \text{ in } F(X), \quad \epsilon_v \in \{-1, 1\}. \quad (6.1)$$

Here the right side means existence of $m \geq 0$, $h_v \in F(X)$, $r_{j_v} \in R$ and the indicated signs. A specified identity is checked by free reduction.

Lemma 6.1. *Let $H = \langle y_1, \dots, y_\ell \mid S \rangle$ and $G = \langle x_1, \dots, x_n \mid R \rangle$. Then*

$$\begin{aligned}
 \exists f : H \twoheadrightarrow G & \iff \begin{aligned} & \exists u_1, \dots, u_\ell \in F(X), v_1, \dots, v_n \in F(Y) : \\ & s(u_1, \dots, u_\ell) \in \langle\langle R \rangle\rangle \quad (s \in S), \\ & v_j(u_1, \dots, u_\ell) x_j^{-1} \in \langle\langle R \rangle\rangle \quad (1 \leq j \leq n). \end{aligned}
 \end{aligned}$$

Together with products (6.1) for these finitely many normal-closure memberships, the words u_i, v_j have a primitive-recursive checking predicate.

Proof. The first family of identities induces

$$f : H \longrightarrow G, \quad y_i \longmapsto u_i \langle\langle R \rangle\rangle.$$

The second gives $f(v_j) = x_j \langle\langle R \rangle\rangle$, hence $\text{im } f = G$. Conversely, choose words representing the images and preimages of the generators under an epimorphism. Its defining identities lie in $\langle\langle R \rangle\rangle$ and therefore admit (6.1). Substitution and free reduction of finite words, and the finite integer and index checks in (6.1), are primitive recursive. $\square$

Proof of Theorem 1.1. Write $\mathbf{A}(d, r)$ for the margin condition (3.2) on finite Laurent data d , and $P_{d,r}$ for the presentation (4.1)–(4.4) at the radius (3.4). Let $\mathbf{E}(H, P; \eta)$ check the words and products in Lemma 6.1. For a decoded witness $c = (d, r, \eta)$, set

$$V(P, c) = 1 \iff \mathbf{WF}(P) \wedge (n(P) = 0 \vee [\mathbf{A}(d, r) \wedge \mathbf{E}(P_{d,r}, P; \eta)]), \quad (6.2)$$

where $\mathbf{WF}$ checks the generating alphabet and $n(P)$ is its size. All finite syntax checks are included in $\mathbf{A}$ and $\mathbf{E}$. The radius is an integer expression in the data, and

$$\{v \in \mathbb{Z}^k : \|v\| < R\} = \left\{ v \in [-R, R]^k \cap \mathbb{Z}^k : \sum_i v_i^2 < R^2 \right\}.$$

Consequently, constructing $P_{d,r}$ uses a bounded finite list. Lemma 3.1 and Lemma 6.1 show that V is primitive recursive.

For $n(P) > 0$, soundness is the implication

$$V(P, c) = 1 \implies P''_{d,r} = 1, \quad P_{d,r} \twoheadrightarrow G(P) \implies G(P)'' = 1,$$

by Proposition 4.1. For completeness, Proposition 5.2 and Lemma 3.1 give

$$G(P)'' = 1 \implies \exists d, r : \mathbf{A}(d, r), \quad P_{d,r} \twoheadrightarrow G(P).$$

Lemma 6.1 supplies η . If $n(P) = 0$ and P is well formed, $G(P) = 1$, as required by the first branch of (6.2). This proves (1.1). Finally, projection of the primitive-recursive relation $\{(P, c) : V(P, c) = 1\}$ gives the asserted recursive enumeration. $\square$

Foundation model assistance for Formalisation

Theorem 1.1 and its algebraic ingredients are formalised in Lean 4 with assistance from GPT-Astra through a custom LangGraph harness.

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